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A Minimum Squared-Error Framework for Sampling and Reconstruction in Arbitrary Spaces

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Abstract

We consider non-ideal sampling and reconstruction schemes in which the sampling and reconstruction spaces as well as the input signal can be arbitrary. To obtain a good reconstruction of the signal in the reconstruction space from arbitrary samples, we suggest processing the samples prior to reconstruction with a linear transformation that is designed to minimize the worst-case squared-norm error between the reconstructed signal, and the best possible (but usually unattainable) approximation of the signal in the reconstruction space. If the input signal is known to lie in an appropriately chosen subspace, then we propose a linear transformation that achieves the minimal squared-error norm approximation. We show both theoretically and through simulations that if the input signal does not lie in the reconstruction space, then the suggested methods can outperform the consistent reconstruction method previously proposed for this problem.

1 Introduction

The most common setting considered in the sampling literature is that introduced by Shannon's sampling theorem, in which the input signal is assumed to be bandlimited, the samples of the signal are ideal, *i.e.,* they are equal to the signal values at a set of sampling points, and the reconstructed signal is also a bandlimited function, created by using the sinc interpolation kernel. In practice, however, the input signal is never perfectly bandlimited, and the sampling process is never ideal. Another drawback of the Shannon paradigm is the difficulty in implementing the infinite sinc interpolating kernel, which has slow decay.

To overcome these limitations of the traditional sampling framework, a more recent approach to sampling is to consider samples that can be represented as the inner products of the input signal x with a set of sampling vectors, which form a possibly overcomplete basis (frame) for the sampling space, denoted by $\mathcal{S}$ [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]. Examples include multiresolution decompositions [2, 11], and spline

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